# TEJASWI PADIGALA

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## EDUCATION

**University of Massachusetts Isenberg School of Management, Amherst**

*Master of Science in Business Analytics* Candidate, Dec 2025

- *Relevant coursework*: Business Intelligence & Analytics, Data Management for Business Leaders, Business Application Development in Python, Statistics for Business, Data Science for Business, Project Management, Supply Chain Analytics, Mastering Agile Scrum and Machine Learning for Analytics.

**Guru Nanak Institutions Technical Campus, Hyderabad**

*Bachelor of Technology in Information Technology with Minor in AI & ML* July 2023

- *Relevant coursework:* Big Data Analytics, Database Management Systems, Software Engineering, Data Structures and Algorithms, Object-Oriented Programming, Machine Learning, Computer Networks, Data Mining, Information Security, Artificial Intelligence, Natural Language Processing and Operating systems.

## PROJECT EXPERIENCE

**Analysis of User Reviews on E-Commerce Website**

- Utilized sentiment analysis to categorize user reviews on an e-commerce website as positive, negative, or neutral, demonstrating proficiency in natural language processing techniques.
- Conducted an in-depth analysis of user sentiment to enhance trust between users in e-commerce systems, highlighting the practical application of sentiment analysis in improving user experiences.
- Implemented a classification system to automatically categorize user reviews based on the emotional content of the text, showcasing strong analytical skills and attention to detail.

**Feasibility of Bone Fracture Detection Using Microwave Imaging**

- Conducted research to assess the viability of microwave imaging technology for detecting bone fractures, utilizing theoretical knowledge and practical experimentation.
- Designed and implemented experiments to evaluate the accuracy and effectiveness of microwave imaging compared to traditional imaging techniques like X-rays.
- Analyzed data and presented findings, demonstrating the potential of microwave imaging as a non-invasive and cost-effective method for bone fracture detection.

**Prediction of Crop Yield Using Machine Learning Techniques**

- Developed and trained machine learning models to predict crop yield based on historical data, weather patterns, soil quality, and other relevant factors, demonstrating a deep understanding of agricultural data analysis.
- Implemented feature engineering techniques to preprocess and extract meaningful information from raw agricultural data, improving the accuracy and performance of the predictive models.
- Evaluated the performance of the machine learning models using metrics such as accuracy, precision, recall, and F1-score, showcasing the ability to assess and validate model predictions in real-world scenarios.

**Drone Hub Location for the Prime Air Program**

- Developed an Optimization Strategy for Drone Hub Locations, Conducted a comprehensive analysis of e-commerce trends, demographic data, and drone technology to propose optimal drone hub locations across Massachusetts for efficient last-mile delivery.
- Utilized Data-Driven Decision Making for Deployment Planning, Evaluated factors such as package weight, demand density, and infrastructure to prioritize hub locations in Boston, Worcester, and Springfield/Amherst, ensuring maximum coverage and cost efficiency
- Addressed Operational and Regulatory Challenges, Assessed FAA drone regulations, battery limitations, and weather constraints while designing a phased deployment plan, balancing feasibility, scalability, and compliance requirements.

**Diabetes Prediction**

- Developed Machine Learning Models for Diabetes Prediction, Implemented and evaluated models such as Random Forest, Logistic Regression, Gradient Boosting, and KNN to classify diabetes risk with ~90% accuracy.
- Performed Data Analysis and Feature Engineering, Conducted exploratory data analysis (EDA) to identify key risk factors such as BMI, age, and smoking, ensuring optimal feature selection for predictive accuracy.
- Optimized Model Performance and Addressed Challenges, Applied hyperparameter tuning (Grid Search), managed class imbalances, and enhanced prediction reliability by minimizing false negatives for high-risk individuals.

## EMPLOYMENT

**HCL Technologies Limited** Bangalore, KA India

*Graduate Engineer Trainee* Aug 2023-Dec 2023

- Completed comprehensive training in Structured Query Language, Cloud Computing to enhance proficiency in Database Management, software development, Computer networking.
- Contributed to cross-functional project teams, assisting in the development and implementation of software solutions, system upgrades while adhering to project timelines and quality standards.
- Collaborated with team members to troubleshoot technical issues, sharing insights and best practices to improve team performance and achieve project milestones.

## ACTIVITIES

**Member of innovation and research** – *Street cause, India* 2019-2023

- The unit aims to bring about changes in the society by developing and advancing new ideas while implementing new processes.
- It is also the home to all those problems that could be converted into ideas leading to a potential solution.

**Member of Voice 4 Girls Foundation** - *Hyderabad, India* 2021-2023

- Mentor marginalized adolescents in teaching them life- skills through activity-based camps.
- Counceller for training sessions, outstation camps and debrief sessions.

**Campus Residential Student President,** *GNITC, Hyderabad, India* 2020-2022

- Student representant for the residential student community.

## TECHNICAL PROFICIENCY

DBMS, Tableau, MS Advanced Excel, PowerPoint, Word, MS Project, HTML and CSS

*Programming*: SQL, Python, C Lang